

HARDWARE WORKSHOP · STUDENT LAB GUIDE

Inside a Desktop PC

A Disassembly and Reassembly Guide

TARGET	Standard ATX and Micro-ATX Desktop Towers
SCOPE	Full component breakdown, teardown safety, and step-by-step rebuild order
PREREQUISITES	Basic hand-tool safety and anti-static handling fundamentals
LAB FORMAT	Hands-on bench teardown with documented visual checkpoints

Technical Education Series · Department of Computer Systems

Made with GAMMA

ARCHITECTURE

Main components inside the cabinet

Twelve core assemblies establish the computing, power, and thermal foundation of every standard tower.

Motherboard — Main circuit board that routes power and data between all hardware. Connection: ATX standoffs to chassis ground.

CPU — Central processor executing instructions and calculating core logic. Connection: LGA or PGA socket under tension arm.

CPU Cooler & Fan — Draws thermal energy away from the processor heat spreader. Connection: 4-pin PWM CPU_FAN header.

RAM Modules — High-speed volatile memory holding active system data for the CPU. Connection: Keyed DIMM slots with latch clips.

Storage (SSD / HDD) — Non-volatile long-term storage housing the OS, apps, and user data. Connection: SATA data port or M.2 NVMe slot.

Power Supply (PSU) — Converts high-voltage AC mains into regulated DC rails (+12V, +5V, +3.3V). Connection: 24-pin ATX, 8-pin EPS, PCIe plugs.

GPU (Graphics Card) — Dedicated expansion processor rendering visual frames and display feeds. Connection: Primary PCIe x16 slot + rear bracket.

Optical Drive — Optional reader and writer for CD, DVD, or Blu-ray optical media discs. Connection: 5.25-inch chassis bay via SATA.

Case Fans — Maintains directional airflow across the chassis intake and exhaust paths. Connection: 3-pin or 4-pin SYS_FAN headers.

Front-Panel Headers — Connects chassis power buttons, reset switches, activity LEDs, and audio. Connection: Front-panel block (JFP1 header pins).

Internal Cables — SATA lines and PSU harnesses transferring operational power and data. Connection: Drive ports and board receptacles.

Expansion Cards — Add-in cards providing dedicated sound processing or Wi-Fi connectivity. Connection: PCIe x1 or secondary PCIe x4 slots.

Made with GAMMA

EQUIPMENT

Tools and safety checklist

Gathering the right bench kit prevents damaged fasteners, broken sockets, and stripped threads.

Phillips #2 Screwdriver

Drives standard case thumbscrews, expansion brackets, and drive mounting bays without stripping heads.

Anti-Static Wrist Strap

Grounds your body continuously to protect delicate integrated silicon circuits from electrostatic discharge.

Magnetic Parts Tray

Separates board standoffs, drive screws, and case panel bolts by thread type and length during work.

Velcro Straps & Cable Ties

Bundles wire harnesses flush against the backplate to ensure unobstructed internal airflow paths.

Thermal Compound & Alcohol

Restores microscopic contact between the CPU lid and cooler base whenever the heat sink is lifted.

Inspection Light & Camera

Illuminates tiny pin markings and records exact motherboard routing before any plug is disconnected.

BENCH RULE

Never substitute magnetic power drivers for manual hand screwdrivers when mounting circuit boards into case standoffs.

Made with GAMMA

PROTOCOLS

Safety precautions before you start

Observe these eight core lab protocols to safeguard both technician safety and system silicon.

Unplug Mains & Hold Power Button

Cut the AC supply and hold the chassis button for 10 seconds to drain remaining capacitor energy.

Switch Off the Power Supply Rocker

Set the I/O rocker switch on the rear PSU plate to the O position to ensure absolute electrical isolation.

Work on a Dry, Non-Carpeted Surface

Conduct bench work exclusively on wooden or anti-static mats to prevent friction-generated charge build-up.

Ground to Bare Chassis Metal

Clip your wrist strap to unpainted interior frame steel or touch unpainted metal before handling boards.

Handle Boards by Outer Edges Only

Avoid touching surface-mount solder traces, gold PCIe contact fingers, or delicate component pins directly.

Never Open the Power Supply Enclosure

Internal high-voltage capacitors inside the PSU retain lethal charges long after the unit has been unplugged.

Keep Liquids Away from the Workspace

Maintain an absolute exclusion zone for drinks and moisture anywhere near active hardware teardown tables.

Photograph Layouts & Sort Fasteners

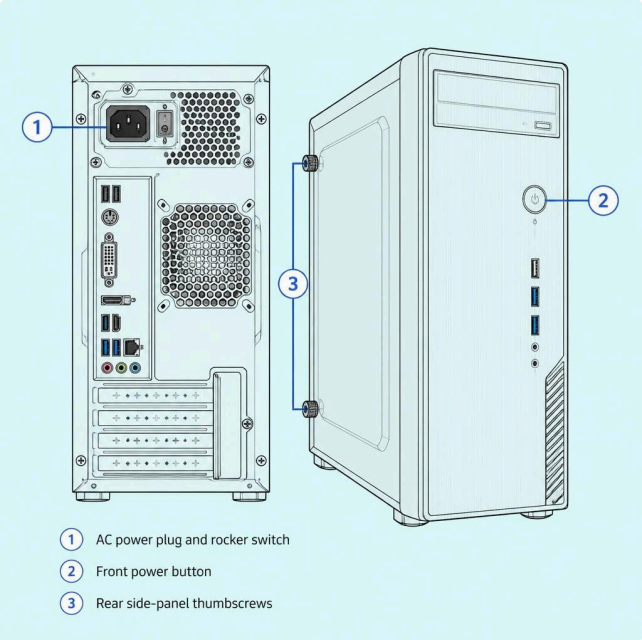
Document cable header groupings prior to disconnection and partition screws into marked bins immediately.

Made with GAMMA

DISASSEMBLY · STAGE 1

Disassembly steps 1–3: power down and open the case

Isolating system power, neutralizing board capacitors, and sliding chassis panels.



Technical diagram: Steps 1, 2, and 3 isolation sequence.

1 *Step 1: Isolate AC Power*

Flip the rear PSU rocker switch to 0 (off), unplug the wall power cord, and detach all peripheral cables including monitors, mice, and keyboards.

2 *Step 2: Drain Residual Energy*

Press and hold the front power button for 10 full seconds to discharge power supply capacitors and avoid short circuits while touching board pins.

3 *Step 3: Remove Side Panel*

Loosen the two rear thumbscrews securing the primary side panel, slide the metal sheet rearward, and capture a reference photo of the full harness layout.

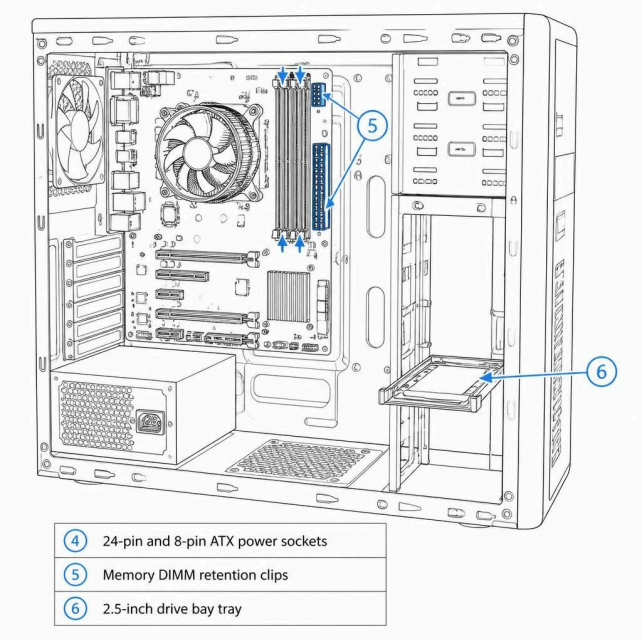
Note: Technical illustrations are used across disassembly cards to show precise access points without visual clutter.

Made with GAMMA

DISASSEMBLY · STAGE 2

Disassembly steps 4–6: disconnect cables, remove RAM and storage

Releasing motherboard harnesses, disengaging memory latches, and sliding out drive trays.



Technical diagram: Steps 4, 5, and 6 harness and module release.

4 *Step 4: Unplug Cables*

Depress retention clips to pull the 24-pin main ATX plug, 8-pin CPU power cable, SATA lines, front-panel wires, and fan headers from the motherboard.

5 *Step 5: Extract RAM Sticks*

Press down firmly on the plastic locking latches at both ends of each DIMM channel until the RAM pops upward, then lift modules out by their bare edges.

6 *Step 6: Remove Storage*

Unthread the side mounting screws or squeeze tool-less bay brackets to slide the SSD or mechanical hard drive out of its drive cage smoothly.

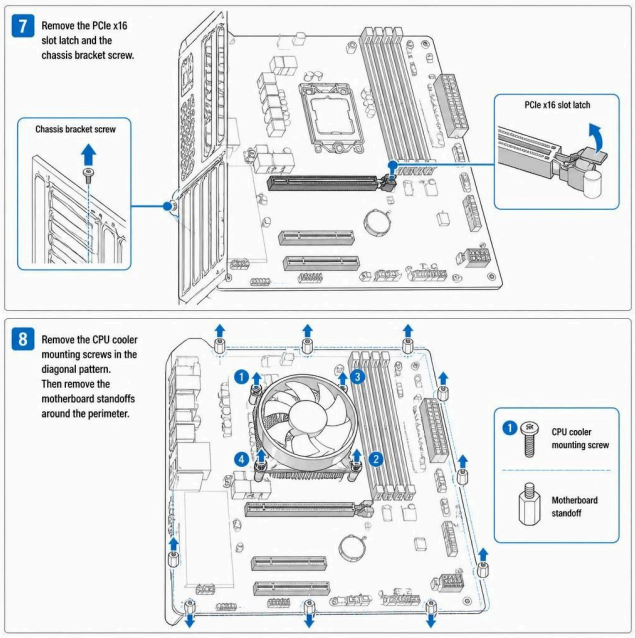
Tip: Never pull cables by their flexible wires; always grip the hard plastic molded connector body directly.

Made with GAMMA

DISASSEMBLY · STAGE 3

Disassembly steps 7–8: remove the graphics card, cooler and motherboard

Extracting PCIe expansion cards, unmounting thermal hardware, and lifting the main PCB.



Technical diagram: Steps 7 and 8 board extraction points.

7 *Step 7: Remove Expansion Cards*

Unscrew the metal chassis rear bracket. Press the plastic PCIe slot latch down until it clicks open, then gently lift the graphics card vertically out of its socket. Remove any other expansion cards.

8 *Step 8: Remove Cooler & Board*

If needed, unscrew the PSU. Gently remove the CPU cooler, loosening its screws diagonally in cross-pattern quarter turns to ease tension. Undo all motherboard perimeter screws, noting standoff locations, and lift the board clear.

STANDOFF CAUTION

Account for every metal brass standoff threaded into the case tray. A loose or misplaced standoff under the motherboard will bridge electrical traces and cause catastrophic shorts.

Made with GAMMA

REASSEMBLY · PHASE 1

Reassembly steps 1–3: board, CPU area and memory

Building outward from the motherboard foundation to keep unobstructed access to delicate components.

1 *Step 1: Verify I/O Shield & Standoff Alignment*

Alignment & Foundation

Snap the rear I/O shield firmly into the case frame from the inside until all edges seat flush. Verify that brass motherboard standoffs in the chassis tray match your exact board mounting holes.

2 *Step 2: Seat & Fasten the Motherboard*

Even Mechanical Mounting

Lower the motherboard at an angle into the I/O shield ports, lower it onto the standoffs, and tighten center and corner screws gradually in a cross pattern without overtightening.

3 *Step 3: Apply Paste, Mount Cooler & Insert RAM*

Thermal Layer & Memory

Clean surfaces with alcohol, place a pea-sized dot of thermal paste on the CPU, and tighten cooler brackets evenly. Push RAM into keyed channels until both retention clips snap shut.

WHY THE ORDER MATTERS

Mounting the CPU cooler and memory modules while the mainboard area has total clearance prevents your hands from knocking against chassis bays, power harnesses, or drive cages.

Made with GAMMA

REASSEMBLY · PHASE 2

Reassembly steps 4–6: storage, cards, power and cables

Mounting secondary drives, locking graphics cards, and dressing power harnesses cleanly.

4 *Step 4: Mount Hard Drives & SSDs*

Storage Bays

Slide solid-state and hard drives into internal bays, orienting their connection ports toward the cable routing cutouts, and secure them firmly using dedicated drive screws.

5 *Step 5: Lock GPU & Expansion Cards*

PCIe Slots

Align the graphics card with the primary PCIe x16 slot, press down until the slot lock engages with an audible click, and fasten the metal rear bracket screw to the chassis.

6 *Step 6: Route & Connect All Cables*

Power & Header Distribution

Attach 24-pin ATX, 8-pin CPU, PCIe power, SATA cables, front-panel pins, and case fans. Route bundles behind the tray with ties to keep internal airflow channels completely clear.

WHY THE ORDER MATTERS

Routing bulky PSU lines last ensures cables never get trapped underneath delicate expansion cards or block access to tiny front-panel header pins on the lower edge of the motherboard.

Made with GAMMA

VERIFICATION

Final checks and first boot

Completing pre-power inspection, confirming POST initialization, and diagnosing common startup faults.

PRE-FLIGHT VERIFICATION

Closing the Build

- Inspect the chassis bed for stray screws, clipped ties, or unseated metal tabs.
- Confirm all fan blades rotate freely without snagging nearby wires.
- Slide the side panel shut and secure rear thumbscrews.
- Connect monitor, keyboard, mouse, and AC mains power cord.
- Flip the PSU rocker to 1 and press the front power button.
- Verify successful POST, fan spin-up, and boot display output.

BENCH TROUBLESHOOTING

Diagnostic Solutions

No Power
Check wall outlet, ensure rear PSU rocker is switched to I, and verify the front-panel POWER SW header is seated on the correct motherboard pins.

Fans Spin, No Display
Verify monitor cable connects to the graphics card rather than motherboard video port, and confirm auxiliary PCIe power cables are seated.

Beep Codes / DRAM LED
Power down, release RAM latches, and reseat DIMMs firmly until both clips snap to resolve poor memory contact.

A thorough visual check before connecting power prevents catastrophic short circuits and saves hours of diagnostic bench work.

Made with GAMMA